

IoT-Based Precision Litchi Tracking and Counting Method Using Gated Metrics

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Abstract—Accurate and efficient multiobject tracking and counting methods are designed to address the challenges of counting in complex environments. This study presents a novel tracking and counting method called LitchiCount, integrating the multiobject tracking detection model LitchiDet with a counting module to address issues, such as missing counts, repeated counts, and the lack of interpretability commonly found in traditional machine learning approaches. The method is designed with the guidance of the visual interpretable method Grad-CAM++, as well as the experimental validation method based on important features. To improve the detection accuracy of small targets under dense occlusion and overlapping, we proposed LitchiDet, which combines a small target detection layer, a decoupled fully connected attention with C3Ghost module (DFC-C3Ghost), and an efficient layer aggregation network block (ELANB). Our counting module improves target tracking accuracy and robustness in dense occlusion scenes while reducing counting errors from scene changes. We propose a distance-generalized intersection over union association metric using a gating mechanism (DG-GM) and an AreaC counting strategy tailored to field intricate scenes. Finally, to enhance Internet of Things (IoT) deployment, we migrated LitchiCount to the Jetson AGX Xavier platform and optimized the model with TensorRT, significantly improving computational efficiency and real-time performance, particularly in resource-limited IoT environments, meeting real-time and low-power demands. The results demonstrated that our proposed method outperforms state-of-the-art detection models, as well as DeepSort-based counting methods in detection and counting. Importantly, by applying our method to the scenario of detecting and counting litchi from multiple perspectives in

a field setting, we achieved low-repetitive and reliable counting, demonstrating the robust performance of this approach in real-world applications.

Index Terms—Fruits counting, interpretability, Litchi, multiple object tracking (MOT), object detection, real-time.

I. INTRODUCTION

ACCURATE yield estimation is crucial for improving the economic benefits of orchards. With the rapid development of agricultural Internet of Things (IoT), the integration of IoT-based data acquisition devices with deep learning algorithms has become a promising research trend. This study aims to develop a high-precision and interpretable litchi counting method by combining key agricultural IoT technologies, such as sensors, cameras, and edge computing platforms, to address challenges in data collection, analysis, and processing. By seamlessly integrating these technologies, the method significantly improves counting accuracy and computational efficiency, effectively overcoming common issues such as missing counts, repeated counts [1], and environmental constraints that arise in real-world field conditions.

In recent years, numerous methods have been proposed for fruit detection using machine learning techniques [2]. While the machine learning-based fruit detection studies have demonstrated satisfactory detection performance, detecting fruit in real-world field environments presents significantly more challenging and intricate conditions. This often leads to a significant number of missed detections and false positives. Furthermore, there are relatively few optimizations specifically aimed at deploying on IoT edge devices with limited storage space and computing power, which poses challenges for achieving real-time operation and efficient data processing in resource-constrained IoT environments. A number of regression counting methods for fruit yield estimation, crop counting, etc., have also been proposed [3], [4]. Although these counting methods can provide approximate fruit count results within an acceptable margin of error, they suffer from the loss of location information for each individual fruit target. This limitation hinders the ability to accurately associate the same target and often resulting in missing counting and repeat counting.

On the other hand, the rapid development of agricultural IoT, artificial intelligence, and object tracking technologies has sparked growing interest in utilizing deep learning-based methods for fruit detection and counting [5], [6], [7].

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There are two main methods for fruit detection. The first is one-stage target detection models, such as SSD [8], Yolo series [9], etc. They show strong robustness in night and dense scenes [10], [11], [12]. The second is two-stage target detection models, such as Faster R-CNN, Mask-RCNN and so on [13], [14], [15]. Fruit counting methods primarily include single-view or dual-view detection and counting [16], [17], LIDAR and thermal imaging counting [18], [19], and counting by object tracking [20], [21], [22]. Recent advances in multiobject tracking (MOT) have demonstrated particular effectiveness in handling occlusions. For instance, Viveros Escamilla et al. [59] proposed a fruit counting system based on YOLOv5 for detection and DeepSORT for tracking, achieving 85.7% accuracy in challenging greenhouse conditions through temporal association. Similarly, Feng et al. [60] developed a dynamic counting method using YOLOv7-tiny and an improved Hungarian algorithm, enhancing fruit association accuracy to 81.02% in highly occluded orchards. Other tracking improvements include cascade refinement [23], integration of semantic segmentation modules [24], and optimization of cross-modal fusion [25].

Despite the strong performance demonstrated by the deep learning-based methods, challenges, such as frequent tracking identity (ID) switches, repeat counting, and missed detections still arise in diverse scenarios, particularly when litchi targets are located at the edge of the field of view (FOV) due to limitations in video acquisition. More importantly, the majority of deep learning models exhibit high complexity, lack transparency, and interpretability, which poses challenges for individuals in comprehending the decision-making mechanisms of the models and enhancing their performance [26], [27].

Although existing fruit detection and counting methods have achieved some success, they still face numerous challenges in real agricultural environments. Traditional machine learning methods perform poorly in complex lighting conditions, dense occlusion, and small object detection, leading to high false negative and false positive rates. While current deep learning methods have improved detection accuracy, their model complexity is too high to be deployed for real-time operation on resource-constrained IoT edge devices. Furthermore, existing multiobject tracking methods are prone to issues, such as ID switching and duplicate counting, when targets frequently enter and exit the FOV or exhibit nonlinear motion, and they lack effective target association strategies. These limitations significantly restrict the feasibility and reliability of real-time yield estimation in precision agriculture. Therefore, the development of a lightweight, high-accuracy, and robust real-time litchi counting method is of great theoretical and practical significance for enhancing orchard economic benefits and advancing the practical application of agricultural IoT technologies.

To address the aforementioned issues of fruit detection and counting, We propose a fast video counting method (as shown in Fig. 1) for dense field scenes. Specifically, our main contributions are summarized as follows.

- 1) The first attempt to work on litchi fruit video counting from multiple sides and achieve robust tracking and counting performance.
- 2) We use Grad-CAM++ and the experimental validation method based on important features to identify the reasons for the poor performance of litchi detection in the field and determined the features that contribute more significantly to data association decisions. Based on these findings, we develop a counting method that significantly reduces repeat counting, thereby improving the accuracy and interpretability of litchi counting.
- 3) We present LitchiDet, a lightweight model for litchi detection. The introduction of a small target detection layer significantly improves the detection accuracy of small litchi in dense scenes. The decoupled fully connected attention with the C3Ghost (DFC-C3Ghost) module, which incorporates a hardware-efficient attention mechanism, captures long-range pixel dependencies, thereby enhancing model representation and highlighting fine litchi features. The efficient layer aggregation network block (ELANB) module improves gradient flow and delivers robust performance in scenarios involving occlusion and overlap. Compared to existing fruit detection models, LitchiDet not only achieves higher detection accuracy but also maintains efficient performance on hardware devices.
- 4) A tracking and counting module is designed. Specifically, the distance-generalized IoU based on the gating mechanism (DG-GM) is proposed to significantly reduce the error caused by fruit ID switching and further improve the tracking association [55] accuracy under dense scenes. Then, an area counting strategy (AreaC) based on field litchi scenes is developed, which effectively mitigates the occurrences of missing and repeat counting in video sequences under varying scene.
- 5) The experiments, compared with existing DeepSort-based tracking and counting methods, demonstrate that the proposed approach outperforms in both tracking accuracy and counting reliability. To further enhance real-time processing capabilities, the LitchiCount model has been successfully deployed on the Jetson AGX Xavier platform. By leveraging the TensorRT optimizer, the detection and tracking components have been significantly accelerated, enabling efficient inference while minimizing computational latency. Specifically, the Jetson AGX Xavier functions as the runtime platform, seamlessly executing detection, tracking, and counting tasks in-field. It integrates the LitchiDet module for precise litchi detection and the tracking-counting module for association and trajectory analysis. This architecture allows LitchiCount to achieve high-precision litchi counting directly on edge devices, making it a promising solution for deployment in unmanned vehicles to automate orchard patrol operations.

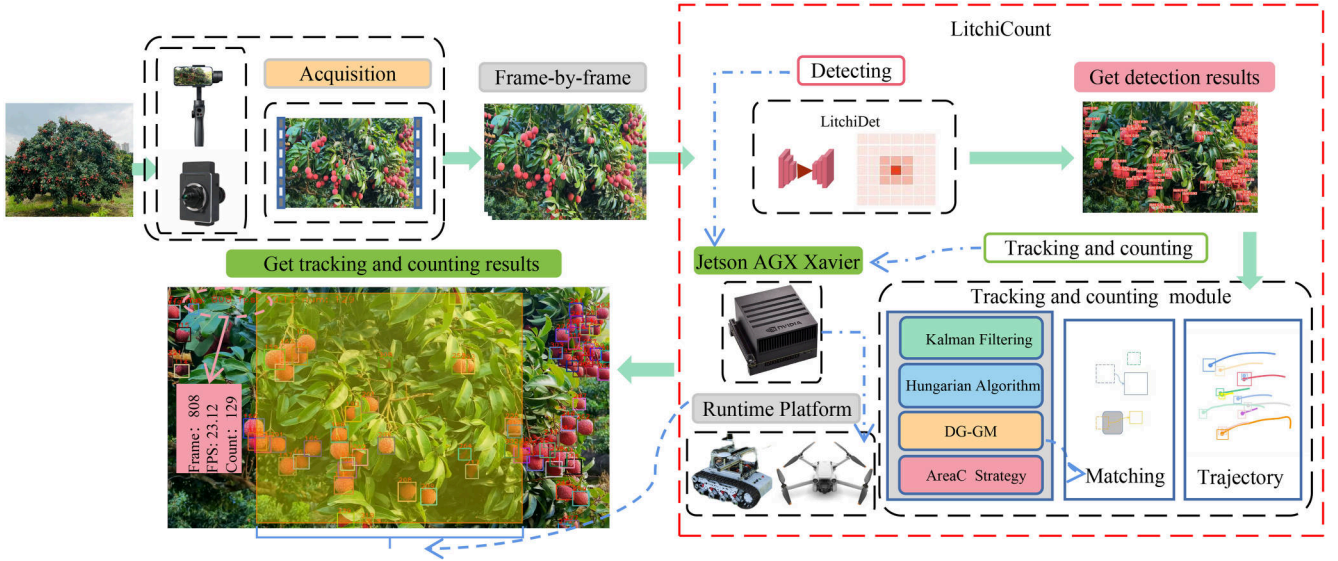


Fig. 1. Framework of LitchiCount.

II. DATASET PREPARATION

A. Data Acquisition

In this section, we present a new dataset for litchi detection and video counting, collected in Conghua and Zengcheng, Guangzhou, China, between June and July 2022–2023. The dataset includes litchi species, such as Huai Zhi, Guiwei, and Xianjinfeng. Data were captured using a Huawei Mate40 (12MP, f/1.9), an iPhone13 (12MP, f/1.6), and a 3-axis gimbal. To improve model accuracy under various conditions, we captured images and videos featuring variations in size, brightness, occlusion, and density from multiple angles at distances of 0.5–2 m. In total, 475 RGB images containing over 23 000 litchi targets were collected, alongside videos totaling 933 s. The movement speed during collection ranged from 0.1 to 0.3 m/s, with a frame rate of 30 FPS and a resolution of 1080p. Fig. 2 provides an overview of the data collection process and sample images.

B. Data Augmentation and Dataset Construction

In the initial dataset construction, we manually annotated litchi instances using Labellmg software. The dataset was split into training (334 images) and test sets (141 images) in a 7:3 ratio for model training and evaluation. Data augmentation techniques, including horizontal flipping, brightness adjustment, motion blur, simulated rainfall, and salt-and-pepper noise, were applied to simulate diverse field conditions and reduce overfitting [28]. This increased the training set to 2004 images. For further validation, the litchi counts in each video sequence were manually precounted.

III. DESIGN FOR INTERPRETABLE WORK

Our proposed method LitchiCount is designed under the guidance of interpretability methods and follows the tracking-by-detection paradigm, with the foundation and core of this paradigm being object detection and data association [29].



Fig. 2. Acquisition process and image data examples.

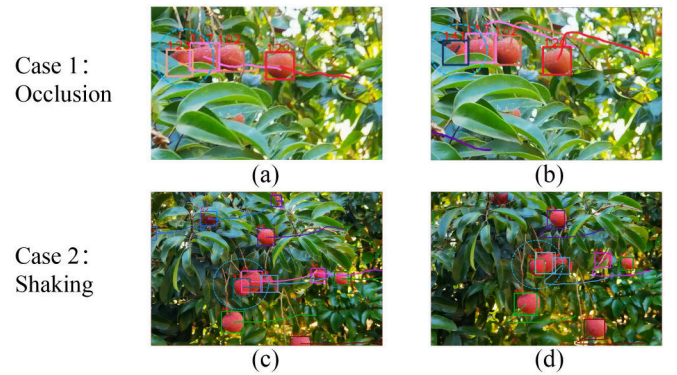


Fig. 3. Example diagram of trajectory interruption. (a) t-frame moment of Case1. (b) t+15 frame moment of Case1. (c) t-frame moment of Case2. (d) t+9 frame moment of Case2.

To enhance the detection performance, we conducted interpretability research using Grad-CAM++ as a method [30]. *Step1*: The output of a convolutional neural network (CNN) for classification is defined as Y . Next, the output score Y for a particular class c is computed by performing a dot product

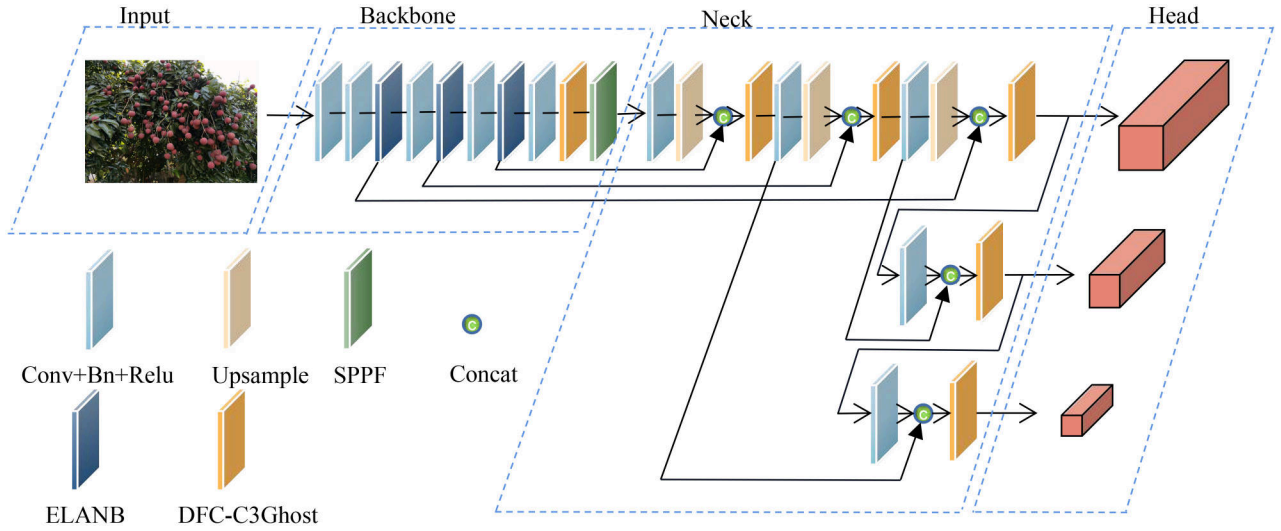


Fig. 4. Structure of LitchiDet.

between the weight W and the feature map K

$$Y^c = \sum_k w_k^c \cdot \sum_i \sum_j A_{ij}^k \quad (1)$$

where w_k^c represents the weight of the k th feature map corresponding to class c , and A_{ij}^k represents the pixel value at a particular spatial location (i, j) within the k th feature map. *Step2:* The heatmaps L_{ij}^c are obtained by applying the principle of weighted summation. Specifically, the weights of feature map are determined by calculating a weighted average of pixel-scale gradients. It is one of the widely used visual explanation methods and explains the predictions of models based on CNNs through image visualization. Specifically, it can accurately localize the predicted class and generate heatmaps at all positions of the class, which highlight the key regions that influence the model's decision-making. The heatmaps are obtained through the following steps:

$$w_k^c = \sum_i \sum_j \alpha_{ij}^{kc} \cdot \text{relu} \left(\frac{\partial Y^c}{\partial A_{ij}^k} \right) \quad (2)$$

$$L_{ij}^c = \sum_k w_k^c \cdot A_{ij}^k \quad (3)$$

where α_{ij}^{kc} represents the gradient weights, which signifies gradient weights for the pixel for class c at position (i, j) in the k th feature map A^k

In this article, we utilize heatmaps generated by Grad-CAM++ to highlight the regions of interest within our litchi detection model. This process aids in identifying issues, designing with a specific target in mind, and making improvements to enhance the model's interpretability.

In multiobject tracking, data association plays a crucial role in matching and associating targets by calculating their similarity between adjacent frames [31], [32]. At present, target similarity is mainly determined by the motion information or appearance features between adjacent frames. According to preliminary experiments, the original tracking model is prone to trajectory interruption under conditions, such as

fruit occlusion and camera shaking, as shown in Fig. 3. Hence, we employ an experimental validation method based on important features to guide the enhancement of the tracking model. In particular, we modify or introduce two crucial features, namely, target motion information and appearance information, into the original tracking model to examine their individual impacts on the model's prediction. The features that exert a significant impact on model predictions and lead to higher accuracy are the ones that contribute more to data association.

Based on this idea, we enhanced the IoU metric and integrated an appearance feature metric into the original tracking model. Additionally, fruit counting experiments will be conducted to develop a scheme that achieves higher performance and better interpretability.

IV. LITCHIDET DETECTION MODEL DESIGN

Considering the existing issues of missed and false detections in fruit detection models, we introduce a lightweight litchi detection model, LitchiDet, designed for edge computing platforms in this section (as shown in Fig. 4). The following is an introduction to the three core components in the LitchiDet.

A. Low Computational Cost Global Attention Method Design

Recently, some scholars have suggested the integration of self-attention mechanisms, like transformers, with ghost modules [33]. Despite their ability to capture global information, these modules require a considerable number of reshape and transpose operations. This poses challenges for hardware-friendly deployment [34], [35]. The GhostNetV2 module [36], which incorporates the DFC and Ghost module, will be a satisfactory solution to the aforementioned issues, so we incorporated it into the C3 module to form DFC-C3ghost.

This method offers the following benefits: 1) it generates more features with fewer parameters, maintaining low-computational cost while improving global information

attention and 2) it reduces the attention module's computational complexity from $\mathcal{O}(H^2W^2)$ to $\mathcal{O}(H^2W + HW^2)$, aiding hardware deployment. The input feature $X \in \mathbb{R}^{H \times W \times C}$ is processed through two branches. The Ghost module generates the output feature, and the DFC attention mechanism computes the attention matrix. DFC attention works on downsampled features, then upsamples them to the original size. DFC's fully connected (FC) layer is split into horizontal and vertical FC layers to prioritize horizontal and vertical pixels in the feature map. This division helps capture long-range relationships in both directions, enhancing Ghost module's output features. After a 1×1 convolution on the input, we get $Z \in \mathbb{R}^{H \times W \times C}$ for the attention module. The long-range spatial information in the horizontal direction of Z is computed as

$$a'_{hw} = \sum_{h'=1}^H F_{h,h'}^H \odot z_{h'w}, h = 1, 2, \dots, H, w = 1, 2, \dots, W. \quad (4)$$

Based on (4), the formula to calculate the long-range spatial information in the vertical direction is as follows:

$$a_{hw} = \sum_{w'=1}^W F_{w,hw'}^W \odot a'_{hw'}, h = 1, 2, \dots, H, w = 1, 2, \dots, W \quad (5)$$

where $\odot$ is the element-wise multiplication of the array, and F represents the learnable weights in the FC layer. The FC layer can eliminate tensor reshaping and transposition operations that significantly affect inference speed by sharing a portion of weights F .

Finally, the output ϑ is obtained by performing element-wise multiplication on the results of the two branches of the GhostnetV2 module

$$\vartheta = \text{Sigmoid}(A) \odot V(X) \quad (6)$$

where $A = \{a_{11}, a_{12}, \dots, a_{HW}\}$ represents the attention map obtained by the DFC attention mechanism, and V represents the output features of the Ghost module.

Therefore, in order to highlight intricate details of litchi fruits in dense scenes and suppressing irrelevant background features, partial C3 modules in the backbone and neck networks were replaced with DFC-C3ghost to capture long-range between pixels at various spatial positions. Consequently, the detection [46] performance of small litchi targets is enhanced while reducing computational costs on edge devices. The network architecture of DFC-C3ghost is depicted in Fig. 5.

B. New Target Detection Scale Design

Our constructed image dataset contains numerous small litchi objects. Nevertheless, the original model, undergoing multiple downsampling operations, has significantly diminished the amount of feature information available for detecting targets smaller than 8 pixels in size [37]. To enhance small litchi detection, we integrated a specialized detection head and incorporated feature fusion layers into the Neck network to leverage both shallow and deep features. The large object detection head was removed due to its minimal contribution,

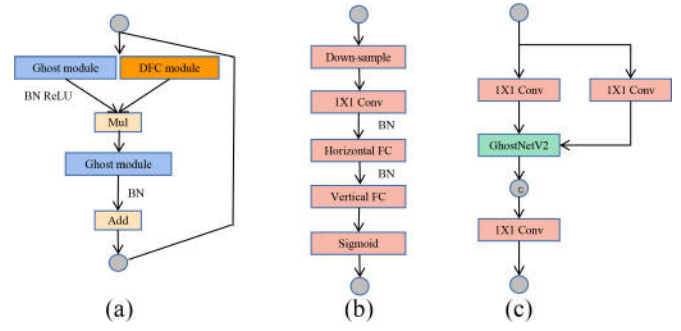


Fig. 5. Architecture of the DFC-C3Ghost module: (a) Basic structure of GhostNetV2, which includes lightweight feature generation using the Ghost module; (b) DFC attention module, which builds long-range dependencies through horizontal and vertical fully connected layers; (c) Complete structure of the DFC-C3Ghost module that integrates GhostNetV2 with the DFC attention mechanism.

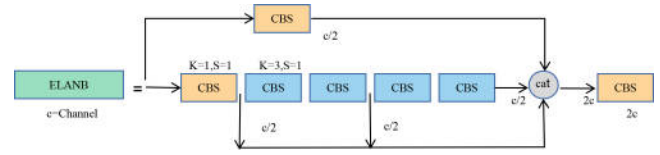


Fig. 6. Structure of ELANB.

allowing the model to focus on small litchi objects and reduce computational costs. The introduction of new feature scales at 160×160 , 80×80 , and 40×40 improves detection accuracy in dense scenes and enhances the extraction of occluded litchi features.

C. Flow Information Enhancement Module Integration

Given the limited feature extraction capability of deep neural networks, researchers have recently proposed network architectures that leverage gradient path design strategies. Unlike traditional mainstream networks that are designed based on data paths, these architectures facilitate the learning of a wider range of features through distinct groups of computational blocks. As a result, deeper networks can effectively learn and converge while maximizing the utilization of parameters and enhancing their generalization capability [38]. In this article, we propose replacing the original C3 module with ELANB (as shown in Fig. 6), a new backbone network structure based on gradient path design strategy. The enhanced network can capture more abundant gradient flow information without introducing additional complex architectures. As a result, this further improves the capabilities of feature information extraction and learning, effectively solving the issue of inadequate extraction of litchi feature information by deep networks in natural environments.

V. LITCHI ONLINE TRACKING AND COUNTING METHOD DESIGN

Given the specific litchi scenarios observed during the experiments, such as dense occlusion, target disappearance, and nonlinear motion, significant challenges arise for existing multiobject trackers based on simple motion models. In response, we referred to ByteTrack [39] and subsequently developed a counting-by-tracking module applicable to litchi

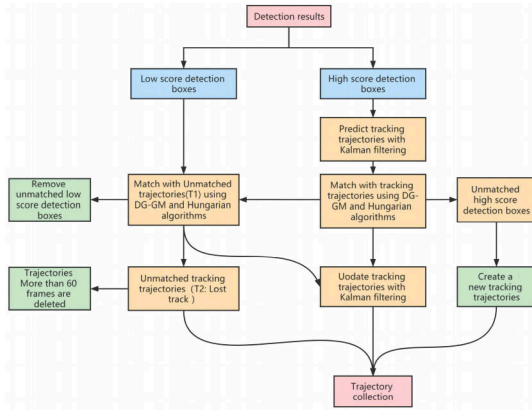


Fig. 7. Flowchart of LitchiCount.

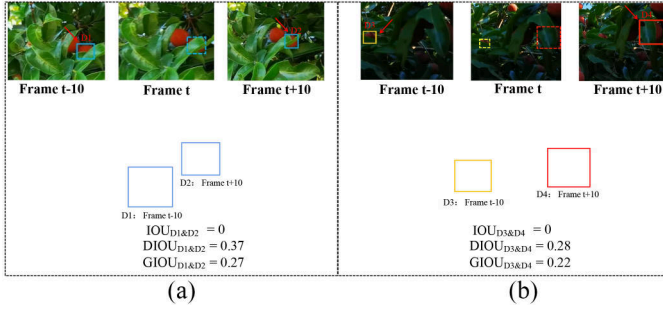


Fig. 8. Comparison of IoU and DG-GM in occlusion and matching range scenarios. (a) Occlusion case: IoU fails to associate detection boxes D1 and D2 (same fruit) due to zero overlap, while DG-GM (via GIOU and DIOU gating) retains spatial relationships for correct association. (b) Matching range expansion: Unconstrained GIOU (yellow box D3 and red box D4) may cause ID drift, but DG-GM uses DIOU to limit distance, preventing misassociation of distinct fruits.

in the field, and fused it with Litchidet to form LitchiCount. We will introduce the details of the mentioned above module and the corresponding implementation details in this section.

A. Overall Implementation

In this section, we mainly describe the implementation process of LitchiCount, the flowchart is shown in Fig. 7. First, the litchi detection results derived from LitchiDet are used as inputs to the tracking module and are divided into high and low-score detection boxes. Subsequently, initial trajectories are created for all the targets in the first frame of the video, followed by three stages of tracking. LitchiCount is specifically designed for video-based tracking and counting. It processes video in a frame-by-frame manner, where each frame is first processed by the detection module (LitchiDet), and then the detected objects are associated using the tracking and counting module. This enables continuous tracking of litchi across frames, ensuring accurate and robust counting in dynamic orchard environments. The first step involves matching using the high-score boxes and all previously generated trajectories. The second step is to use the low-score boxes to match with the trajectories that were not matched in the first step. The third step involves creating a new trajectory for the unmatched high-score boxes from the previous two steps. We retain only 60 frames for the unmatched trajectories and match them when they reappear. Specifically, the Kalman filter is initially applied

to accomplish cross-frame tracking of the fruit in the video. It uses observed data to optimally estimate the system state and aids tracking primarily by predicting and updating target trajectories. That is, the posteriori state vector $\tilde{X}_{m-1}$ of the frame $m-1$ is first used to obtain the priori state vector $\tilde{X}_m$ of the frame m by dot product it with the state transfer matrix F

$$\tilde{X}_m = F\tilde{X}_{m-1}. \quad (7)$$

The prior state estimation covariance P_m^- of the frame m can be represented as

$$P_m^- = FP_{m-1}F^T + Q_{m-1} \quad (8)$$

where Q_{m-1} represents the system noise covariance of the $m-1$ frame.

Updating means using the detection results of the current frame to correct the tracking state associated with it [40]. Here, it is necessary to first calculate the average error u_{m-1} between detection and tracking based on the detection result Z_{m-1} of $m+1$ frames

$$u_{m+1} = Z_{m+1} - H\tilde{X}_m \quad (9)$$

where H denotes the transformation matrix, which is responsible for converting state variables into observation variables.

To obtain the innovation covariance M , it is necessary to map the covariance matrix to the detection space using the transformation matrix H and incorporate the observation noise R

$$M = HP_mH^T + R. \quad (10)$$

Based on (10), the Kalman gain can be expressed as

$$K_{m+1} = P_mH^TM^{-1}. \quad (11)$$

With the above steps, the state update is completed, and the posterior state estimation $\tilde{X}_{m+1}$ and covariance matrix P_{m+1} of the updated current state are represented as follows:

$$\tilde{X}_{m+1} = \tilde{X}_m + K_{m+1}u_{m+1} \quad (12)$$

$$P_{m+1} = (I - K_{m+1}H)P_m \quad (13)$$

where I is the unit matrix.

After the Kalman filtering is accomplished [57], the similarity between the prediction and detection boxes is then calculated using our proposed DG-GM, and the matching is completed by the Hungarian algorithm. At this moment, an ID is assigned to each litchi target for association, and as long as the targets are consistently tracked, their assigned IDs will remain constant. After that, field litchi counting needs to be combined with our proposed area counting strategy AreaC.

B. DG-GM Association Metrics Design

Due to the acquisition device's susceptibility to shaking from angle changes and obstacle avoidance, capturing field images presents additional challenges. Relying solely on the IoU metric can lead to issues with tracking initialization and trajectory interruptions [56], ultimately resulting in significant accuracy errors. To address this issue, we propose DG-GM, a simple and efficient association metric suitable for real-time

Algorithm 1 DG-GM Association Metrics

Input: Bounding boxes of objects and detections, IoU between objects and detection boxes $I[a*b]$, the gating metrics of DG-GM $\alpha(\alpha=0.34)$

Output: DG-GM

- 1: Initialize gating matrix T
- 2: **for** $i = 1; i < a * b; i++$ **do**
- 3: Obtain DIoU between the two boxes by (16)
- 4: **if** $DIoU < \alpha$ **then**
- 5: Obtain IoU between the two boxes by (14) and use it as the matching score
- 6: **else**
- 7: Obtain GIoU between the two boxes by (15) and use it as the matching score
- 8: **end if**
- 9: Update the gating matrix T based on matching score
- 10: **end for**
- 11: Compute the DG-GM by gating matrix T
- 12: return the DG-GM

scenarios. DG-GM improves association performance under conditions of occlusion and changes in motion speed, reducing fruit ID switching and repeated counting. Specifically, IoU is replaced with GIoU [41], which accounts for both the overlap and the smallest enclosing rectangle. The final GIoU value is constrained between 0 and 1 and is obtained through the following steps.

Calculate the IoU of the two boxes

$$IoU(A, B) = \frac{A \cap B}{A \cup B} \quad (14)$$

where A and B refer to detection box A and B , respectively.

Utilize the calculated IoU to compute the GIoU metric

$$GIoU(A, B) = 1 - IoU(A, B) + \frac{|C| - |A \cup B|}{|C|} \quad (15)$$

where C is the minimum bounding box that contains both A and B .

Compared to IoU, GIoU provides a value even when bounding boxes do not overlap, offering a more accurate measure of similarity. This extends the matching range of litchi targets across frames, improving robustness against occlusion and velocity changes [42], [43]. For instance, during occlusion, the IoU between detection boxes D1 and D2 becomes zero [Fig. 8(a)], resulting in ID reassignment when using IoU-based tracking. In contrast, GIoU preserves the relationship between the boxes, correctly associating them with the same fruit.

In addition, we examine the unregulated expansion of the matching range. The tracker may erroneously assign the same ID to different objects, resulting in an amplification of ID drift. As shown in Fig. 8(b), When the yellow detection box that contains the litchi is occluded at frame t , and if there are no constraints on the matching range, the ID of D3 will shift to D4 after 20 frames. Further, we propose association metric called DG-GM that incorporates a gating mechanism. The DIoU, which exhibits greater sensitivity to centroid distance, is initially employed as a gating control mechanism to restrain the unregulated expansion of the matching range [44]. The

DIoU value is also limited in the range of 0 to 1 and it is calculated as

$$DIoU(A, B) = IoU(A, B) - \frac{\rho^2(A, B)}{c^2} \quad (16)$$

where ρ^2 represents the Euclidean distance between the two boxes, and c represents the diagonal length of the minimum bounding box.

During the DG-GM similarity calculation, if the DIoU falls below the specified threshold value, indicating a significant separation between the two boxes, the IoU will be reintroduced to minimize the misjudgment of identical objects beyond the expanded matching range. Otherwise, the GIoU will be used for calculation. The DG-GM association metrics algorithm is shown in Algorithm 1.

After using DIoU for distance limitation, two litchi that are too far apart are no longer recognized as litchi with the same ID when the matching range is extended [as shown in Fig. 8(b)]. This approach ensures robustness in handling occlusion and nonlinear motion scenarios, including velocity changes, while effectively reducing ID drift and confusion, which is extremely critical for accurate litchi counting in large scenes.

C. Area Counting Strategic Develop

In general, litchi counting can be accomplished by tracking unique IDs. However, due to the large size of litchi trees and the limitations of video capture, dynamic adjustment of the FOV becomes necessary, leading to more complex scenarios. Methods such as cross-line or full-area counting may result in significant undercounting or double counting. After a thorough analysis, we identified the following situations.

- 1) When shooting around a litchi tree, it is common for fruits located on one side of the tree to reemerge in the FOV at another angle, usually at the left and right edges of the video. As shown in Fig. 9(a) and (b), all the litchi in the yellow circle have been updated with their IDs after a few frames, which will result in repeat counting.
- 2) Modify the shooting FOV upward or downward to ensure coverage of litchi located at varying heights. As shown in Fig. 9(e), litchi at the lower edge are in an unstable state and tend to switch IDs as the device moves up and down.
- 3) During the camera movement, litchi may be in an unstable state due to occlusion and other conditions. As shown in Fig. 9(c) and (d), the litchi with IDs 179 and 187 in the yellow circle in Fig. 9(c) are detectable, but they have been completely obscured in Fig. 9(d), resulting in a failure to cross the yellow counting line and causing a significant number of missed counts.

To alleviate the above problem, we developed a specialized area counting strategy, named AreaC. As illustrated in Fig. 9(f), the counting region is centrally located within the video frame, covering 60% of the horizontal width and 95% of the vertical height. Litchi are counted based on their unique IDs as they enter this region, where their state is more stable, minimizing the frequency of ID switching. AreaC is designed to maximize the effective counting area while minimizing

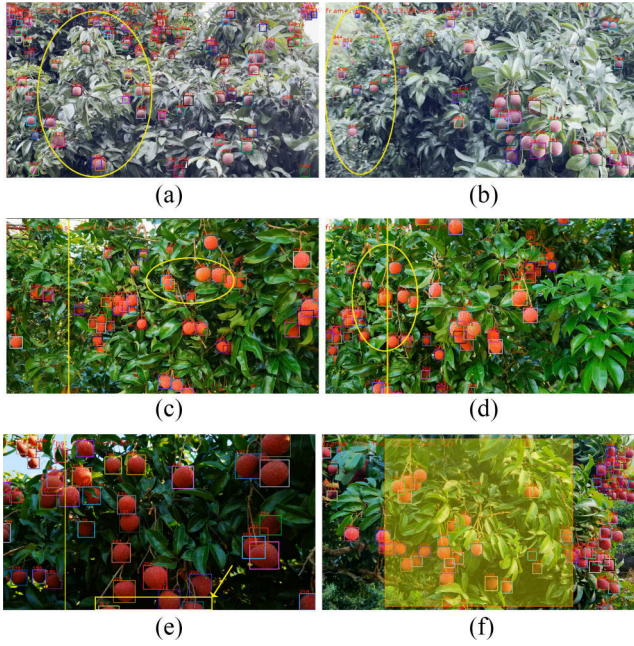


Fig. 9. Scenarios using different counting strategies. (a) and (b) 702-frame and 865-frame picture, respectively, obtained using the Full Count strategy. (c)–(e) 476-frame, the 629-frame, and the anomaly picture which litchi locate at the lower edge of the video, respectively. The yellow line is the counting line. (f) Example of using AreaC for counting, and the yellow area in the diagram is the counting area.

repeated counts, thereby improving the overall efficiency of video analysis. This approach demonstrates superior adaptability to different scene variations and offers higher accuracy compared to the Cross Line and Full Count methods.

VI. TEST EQUIPMENT AND RELATED CONFIGURATION

The experiments in this article are mainly performed by servers and AI edge devices. The server side is mainly for model training and pretest simulation, with platform environment: Windows 10, Intel Xeon Gold 5118 CPU, 128 GB RAM, NVIDIA GeForce RTX 2080(8 GB) GPU, CUDA11.3, PyTorch1.10.0, and Python3.8. The training parameters of LitchiDet are: the input size is 640×640, the learning rate is 0.01, the batch size is 8, the epoch is 200. Nvidia Jetson AGX Xavier is utilized on the edge device side, with the device environment: 8-core NVIDIA Carmel ARMv8.2 64-bit CPU, GPU with 32TOPS. TensorRT, a high-performance deep learning inference optimizer, which compared to pruning [52], distillation [53], and transfer learning [54], can reduce model inference time while maintaining high detection accuracy, is used for hardware-accelerated optimization, mainly for model testing and applications. It optimizes the network structure mainly through model compression and hardware mapping [45]. The LitchiDet detection model is quantized using TensorRT in order to construct a model in TRT format to speed up the inference of LitchiCount on edge platforms. The evaluation metrics in this article are mainly divided into two parts, as follows.

1) *Performance Metrics for detection:* In order to scientifically evaluate the performance of litchi detection model in

complex field environments, we used the following metrics: map, model parameters, model size, and FPS. The precision (P), recall rate (R), and average precision (AP) used for computing the mean AP (mAP) are determined by (17), (18), and (19), respectively. The mAP is then determined by (20). TP, FP, and FN represent true positives, false positives, and false negatives, respectively. In this article, the category is denoted by C , where C has a value of 1

$$P = \frac{TP}{TP + FP} \quad (17)$$

$$R = \frac{TP}{TP + FN} \quad (18)$$

$$AP(k) = \int_0^1 P(R) dR \quad (19)$$

$$mAP = \frac{\sum_1^Q AP(k)}{Q}. \quad (20)$$

2) *Performance Metrics for counting:* Mean absolute percentage error (MAPE), mean error (ME), and FPS are used to evaluate the litchi counting performance in video sequences. Counting accuracy increases as the MAPE and ME decrease. The positive and negative of ME can additionally reflect the situations of missing counting and repeat counting. The manual counting is the average obtained by five different individuals manually counting from the video. MAPE, ME are calculated as

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{\text{Method Count}_i - \text{Manual Count}_i}{\text{Manual Count}_i} \right| \quad (21)$$

$$ME = \frac{1}{n} \sum_{i=1}^n \text{Method Count}_i - \text{Manual Count}_i. \quad (22)$$

VII. EXPERIMENTAL RESULTS

A. Litchi Detection Model Experiments

In this section, we first introduce the process of network optimization based on visual interpretability methods, and then conduct relevant evaluations of the detection model.

1) *Grad-CAM++-Based Model Optimization Process:* Here, Grad-CAM++ is applied to the detection model to generate heat maps, enabling thorough analysis and interpretation of the influential feature regions on the model's prediction results. This analysis serves as the foundation for deriving performance improvement strategies. As shown in Fig. 10, the red and blue dots indicate that they can activate and inhibit decision outcomes. We first generate maps for the 17th layer network of the detection model using Grad-CAM++.

Fig. 10(b) shows that without network improvements, the model focuses only on large litchi targets, indicating the need to enhance detection robustness for small targets. Therefore, we designed a new detection scale with a small target detection layer. To address this issue, we designed a small target detection layer. As shown in Fig. 10(c), this layer improves attention to small litchi, but occluded litchi remain challenging, likely due to the network's limited feature extraction under extreme occlusion. To further address this issue, we introduced the ELANB module. As shown in Fig. 10(d),

TABLE I
ABLATION STUDY OF DETECTION ALGORITHM

Model	New detection scale	ELANB	DFC-C3ghost	map(%)	Parameters (M)	Model size (MB)	FPS
1	Disabled	Disabled	Disabled	83.17	7.01	14.3	22.53
2	Enabled	Disabled	Disabled	87.71	5.45	11.6	20.31
3	Enabled	Enabled	Disabled	88.59	5.73	12.1	20.12
4	Enabled	Disabled	Enabled	88.28	4.39	9	20.43
5	Enabled	Enabled	Enabled	89.14	4.67	9.5	20.29

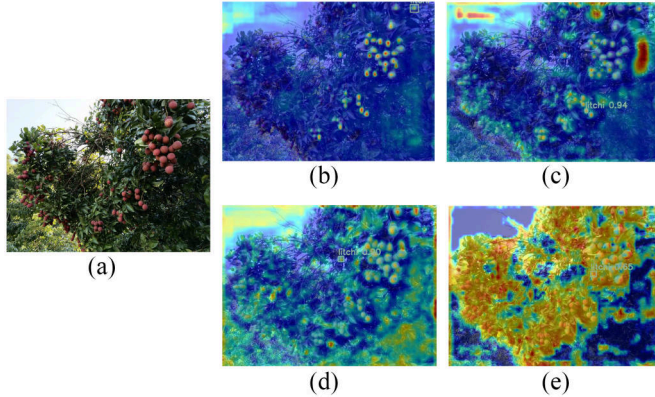


Fig. 10. (a) Heat map of the LitchiDet improvement process. (b) Original detection model, where the attention is primarily on large litchi targets, leading to missed small and occluded fruits. (c) Heat map after incorporating a small object detection layer, improving the focus on small litchi but still struggling with occlusions. (d) Effect of adding the ELANB module, which enhances gradient flow and enables more comprehensive feature extraction, though some irrelevant areas still receive attention. (e) Impact of the DFC-C3Ghost module, which refines attention by suppressing background noise and further strengthening small-object detection.

the improved network focuses on both large and small litchi targets under various occlusion scenarios, while also capturing richer features from dispersed areas like branches, indicating that ELANB enhances feature extraction. However, the increased attention to irrelevant objects raises the risk of false detections. To mitigate this, we added the DFC-C3Ghost attention module, which, as shown in Fig. 10(e), refines the network's focus on litchi regions and enhances multiscale litchi detection. The combination of all three improvements significantly boosts prediction accuracy, the above image visualization results clearly explain how each improvement branch affects the network's decision-making on litchi, which effectively enhances the interpretability and performance of LitchiDet.

2) *Ablation Experiment*: To validate the effect of each improved branch on the LitchiDet, we conducted ablation experiments on our litchi image dataset. From Table I, we observe that the introduction of the small target layer substantially improves the detection model performance. The mAP is also improved by 5.42% and 5.11% with the introduction of ELANB and DFC-C3ghost module, respectively. Although the improved model FPS slightly decreases, the mAP is significantly improved by 5.97% and the model size is compressed by 33.56%. The above proves that the improved LitchiDet can realize high-precision and real-time detection of litchi in large scenarios.

TABLE II
EFFECT COMPARISON OF TENSORRT ON PERFORMANCE AND SPEED

TensorRT	Model	Detection		tracking FPS(AGX)
		map(%)	FPS (AGX)	
×	LitchiDet	89.14	20.30	13.68
✓	LitchiDet	88.89	30.51	17.77

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT OBJECT DETECTION MODELS

Model	map(%)	Parameters (M)	Model size (MB)	FPS
Faster R-CNN	40.72	28.3	108	12.0
Nanodet	40.13	0.95	7.5	32.8
YOLOv6	84.31	18.5	18.5	14.8
YOLOv7	88.98	36.5	74.8	15.3
YOLOv8	87.33	11.1	21.5	17.5
YOLOv11	86.74	13.5	22.0	18.4
EfficientDet	85.50	6.8	15.0	22.5
LitchiDet	89.14	4.67	9.5	20.2

3) *TensorRT Acceleration Comparison Experiment*: To better fulfill the real-time requirements, we adopt the TensorRT technique to improve the edge inference speed of the model. Next, we conducted a comparison experiment. From Table II, it can be clearly seen that after the TensorRT optimization, the model detection accuracy loss is very small, only 0.25%, and the image detection and video counting FPS improvement is significant, about 50.29% and 29.69%. This indicates that TensorRT is significantly useful for LitchiDet, both in maintaining the litchi detection accuracy in dense scenarios and in increasing the operation speed of edge devices with limited storage space and computational power.

4) *Performance Comparison With Other Detection Models*: To further validate the efficiency of the LitchiDet, we conducted performance comparison experiments on Faster R-CNN, Nanodet [47], YOLOv6 [48], YOLOv7 [38], YOLOv8 (As shown in Table III). In terms of detection accuracy, LitchiDet is 118.91% and 122.13% higher than Faster R-CNN and Nanodet and surpasses the above yolo series, in addition, the running speed is also superior to Faster R-CNN and yolo series. The above experiments demonstrate that our proposed LitchiDet is superior to several other models in terms of both performance and speed and enables real-time and efficient recognition of litchi in dense field scenarios.

5) *Performance Comparison of Image Detection After Improvement*: To verify the detection model's improvement, we conducted comparison experiments on litchi images. As shown in Fig. 11, LitchiDet detects more small litchi targets and better distinguishes overlapping litchi in dense



Fig. 11. Detection results of LitchiDet(a) The original model misses small litchi and fails to separate overlapping fruits. (b) Improved model detects small targets, resolves overlaps, and reduces false positives by leveraging multi-scale features, attention mechanisms, and enhanced gradient flow.

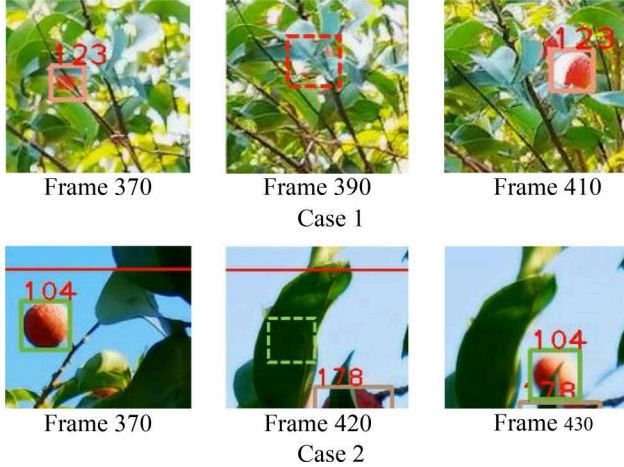


Fig. 12. Examples of keyframe groups that demonstrate performance in the presence of occlusion. The fruits in Case 1 and Case 2 experienced occlusion during the course of the process, yet their respective IDs remained unchanged.

scenes, indicating the improved method significantly enhances detection accuracy.

B. Litchi Counting Experiments

In this section, we perform enhancements to the counting model and evaluate its performance based on the litchi video dataset.

1) *Performance Comparison of Different Metrics and Gating Parameters on Counting:* To validate the applicability of the DG-GM metric, two experiments were conducted. The first compared the performance of various metrics without gating, while the second focused on different gating parameters. In the first experiment, IoU, DIoU, CIoU, EIoU, and GIoU were sequentially integrated into the model as association metrics, followed by counting. Results in Table IV show frequent repeated counting with IoU, while GIoU provided more accurate results. Other methods generally produced lower counts. All approaches improved continuous tracking of litchi occluded by leaves and fruits compared to IoU matching. However, lower counts were attributed to variation in association metrics, leading to an expanded matching range and ID drift between fruits.

Further, in order to select the most suitable gating parameter values to control the matching range, we conducted relevant

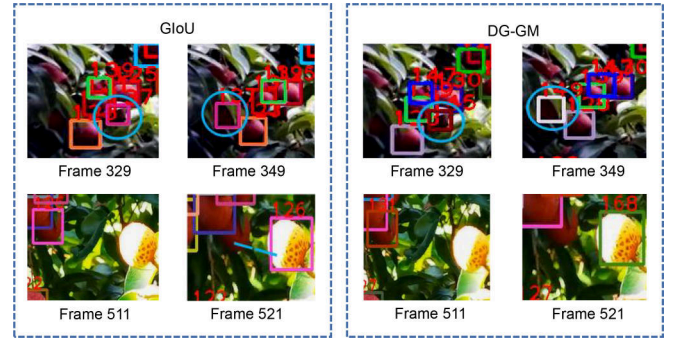


Fig. 13. Examples of keyframe groups that demonstrate performance in the presence of velocity changes. Where DG-GM denotes the improved effect. In Case1, the fruit with ID 137 experienced ID drift, and after improvement, the fruit IDs in the two frames were 145 and 149, and no drift occurred. In Case2, the fruit with ID 126 experienced ID drift, and after improvement, no drift occurred.

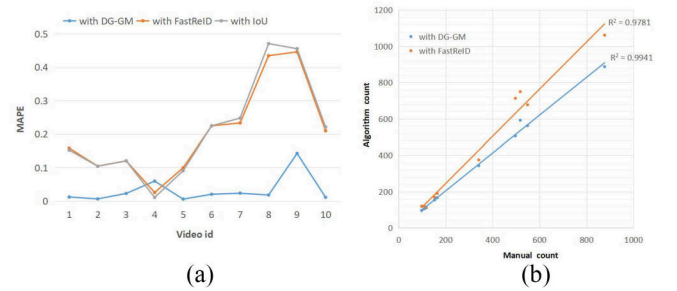


Fig. 14. Performance comparison between DG-GM and FastReID. (a) Comparison of both with the MAPE of the model using IoU before the improvement. (b) Correlation coefficient R^2 between the counting results of the algorithms and the actual results for both.

counting comparison experiments. We selected ten keyframe groups for additional comparison, five of which are to reflect the tracking performance during occlusion, and the other five aim to assess the ability to mitigate ID drift. When the keyframe groups yield positive results, the value increases by 1. Table V demonstrates the counting performance for different gating values. Figs. 12 and 13 show examples of two types of keyframe groups. As shown in Table V, with a gating parameter of 0.34, the keyframe groups all showed a better ability to overcome occlusion and ID drift, and the MAPE was also the lowest. In other words, the DG-GM with this parameter achieves a better balance between continuous tracking ability [58] and addressing the issue of ID drifting when dealing with occlusions, thereby reducing repeat counting.

2) *Experimental Validation Method Based on Important Features:* Additionally, we conducted experiments by integrating the reidentification technique FastReID [49], which allows cross-frame association using appearance information as their similarity. This experiment aimed to compare FastReID with DG-GM and IoU to determine the most effective improvement approach using different important features. As shown in Fig. 14(a), the introduction of FastReID performs similarly to IoU and does not significantly improve the model. In contrast, the DG-GM metric outperforms FastReID in

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT COUNTING
STRATEGIES AND MODELS

Model	Strategy	DG-GM	MAPE(%)	ME
DeepSort(LitchiDet)	Cross Line	×	49.79	-194
DeepSort(LitchiDet)	Full Count	×	46.95	204
LitchiCount	Cross Line	×	51.95	-203
LitchiCount	Cross Line	✓	51.10	-200
LitchiCount	AreaC	×	21.05	90.5
LitchiCount	AreaC	✓	3.29	10

We develop it, which contains a detection model LitchiDet and a tracking and counting module, oriented by Grad-CAM++ and the experimental validation method based on important features. First, LitchiDet which combines a small target detection layer, DFC-C3Ghost, ELANB was proposed. LitchiDet can capture the global information of litchi in complex backgrounds, obtain rich gradient flow information, and then improve the robustness of litchi small targets in scenes with occlusions and overlapping. Second, we propose a tracking and counting module containing an association metric based on a gating mechanism DG-GM and an area counting strategy AreaC. Specifically, DG-GM has the capability to improve the tracking association accuracy and adaptability to velocity variations of litchi under occluded conditions, while AreaC can address the limitations of video acquisition, and finally jointly alleviate the issue of missed counting and repeat counting. Finally, To further address the computational constraints of IoT edge devices, we deployed LitchiCount on the Jetson AGX Xavier platform and leveraged TensorRT for hardware acceleration. The experimental results demonstrate that LitchiCount achieves a MAPE of 3.29%, significantly outperforming DeepSort-based methods in counting performance.

Although the proposed method enables low-repetitive and rapid litchi video counting in the field, certain limitations and areas for improvement remain. The primary challenges are as follows: First, while the method is highly effective for short-term occlusions, it still struggles with long-term occlusion scenarios. Therefore, we plan to further enhance the continuous tracking capability of LitchiCount to handle long-term occlusions and cases where fruits reappear after leaving the frame. Second, when capturing litchi trees surrounded by sparse branches and leaves, fruits on the opposite side may frequently enter the camera's FOV, leading to instances of repeat counting. Consequently, we will continue to explore and incorporate depth information in future work to improve the accuracy of litchi identification and counting at varying distances, thereby reducing the occurrence of repeat counting.

REFERENCES

- [1] F. Gao et al., "A novel apple fruit detection and counting methodology based on deep learning and trunk tracking in modern orchard," *Comput. Electron. Agricult.*, vol. 197, Jun. 2022, Art. no. 107000.
- [2] S. Bargouti and J. Underwood, "Deep fruit detection in orchards," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2017, pp. 3626–3633.
- [3] U. Bhattarai and M. Karkee, "A weakly-supervised approach for flower/fruit counting in apple orchards," *Comput. Ind.*, vol. 138, Jun. 2022, Art. no. 103635.

- [4] S. Madec et al., "Ear density estimation from high resolution RGB imagery using deep learning technique," *Agricult. Forest Meteorol.*, vol. 264, pp. 225–234, Jan. 2019.
- [5] Y. Liu, J. Wang, J. Li, S. Niu, and H. Song, "Machine learning for the detection and identification of Internet of Things devices: A survey," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 298–320, Jan. 2022.
- [6] L. Porzi, M. Hofinger, I. Ruiz, J. Serrat, S. R. Buló, and P. Kontschieder, "Learning multi-object tracking and segmentation from automatic annotations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 6846–6855.
- [7] X. Chang, H. Pan, W. Sun, and H. Gao, "YolTrack: Multitask learning based real-time multiobject tracking and segmentation for autonomous vehicles," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 12, pp. 5323–5333, Dec. 2021.
- [8] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis. ECCV*, 2016, pp. 21–37.
- [9] M. Hniewa and H. Radha, "Integrated multiscale domain adaptive YOLO," *IEEE Trans. Image Process.*, vol. 32, pp. 1857–1867, 2023.
- [10] C. Liang et al., "A visual detection method for nighttime litchi fruits and fruiting stems," *Comput. Electron. Agricult.*, vol. 169, Feb. 2020, Art. no. 105192.
- [11] L. Cui et al., "Context-aware block net for small object detection," *IEEE Trans. Cybern.*, vol. 52, no. 4, pp. 2300–2313, Apr. 2022.
- [12] L. Wang, Y. Zhao, S. Liu, Y. Li, S. Chen, and Y. Lan, "Precision detection of dense plums in orchards using the improved YOLOV4 model," *Front. Plant Sci.*, vol. 13, Mar. 2022, Art. no. 839269.
- [13] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.
- [14] J. Lu et al., "Citrus green fruit detection via improved feature network extraction," *Front. Plant Sci.*, vol. 13, Nov. 2022, Art. no. 946154.
- [15] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2961–2969.
- [16] A. Koirala, K. Walsh, Z. Wang, and C. McCarthy, "Deep learning for real-time fruit detection and orchard fruit load estimation: Benchmarking of mangoyolo," *Precis. Agricult.*, vol. 20, pp. 1107–1135, Feb. 2019.
- [17] L. Wang, Y. Zhao, Z. Xiong, S. Wang, Y. Li, and Y. Lan, "Fast and precise detection of litchi fruits for yield estimation based on the improved YOLOV5 model," *Front. Plant Sci.*, vol. 13, Aug. 2022, Art. no. 965425.
- [18] V. Méndez et al., "In-field estimation of orange number and size by 3D laser scanning," *Agronomy*, vol. 9, no. 12, p. 885, 2019.
- [19] H. Gan, W. S. Lee, V. Alchanatis, and A. Abd-Elrahman, "Active thermal imaging for immature citrus fruit detection," *Biosyst. Eng.*, vol. 198, pp. 291–303, Oct. 2020.
- [20] X. Liu et al., "Robust fruit counting: Combining deep learning, tracking, and structure from motion," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2018, pp. 1045–1052.
- [21] X. Liu et al., "Monocular camera based fruit counting and mapping with semantic data association," *IEEE Robot. Autom. Lett.*, vol. 4, no. 3, pp. 2296–2303, Jul. 2019.
- [22] L. Shen et al., "Real-time tracking and counting of grape clusters in the field based on channel pruning with YOLOV5S," *Comput. Electron. Agricult.*, vol. 206, Mar. 2023, Art. no. 107662.
- [23] S. Ge, C. Zhang, S. Li, D. Zeng, and D. Tao, "Cascaded correlation refinement for robust deep tracking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1276–1288, Mar. 2021.
- [24] X. Xue, Y. Li, X. Yin, C. Shang, T. Peng, and Q. Shen, "Semantic-aware real-time correlation tracking framework for UAV videos," *IEEE Trans. Cybern.*, vol. 52, no. 4, pp. 2418–2429, Apr. 2022.
- [25] C. Li, Z. Xiang, J. Tang, B. Luo, and F. Wang, "RGBT tracking via noise-robust cross-modal ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 5019–5031, Sep. 2022.
- [26] X. Kong and Z. Ge, "Deep PLS: A lightweight deep learning model for interpretable and efficient data analytics," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8923–8937, Nov. 2023.
- [27] W. Dong et al., "Deep spatial-spectral joint-sparse prior encoding network for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 54, no. 12, pp. 7780–7792, Dec. 2024.
- [28] C. Shorten and T. M. Khoshgofaar, "A survey on image data augmentation for deep learning," *J. Big Data*, vol. 6, no. 1, pp. 1–48, 2019.
- [29] Z. Qin, S. Zhou, L. Wang, J. Duan, G. Hua, and W. Tang, "MotionTrack: Learning robust short-term and long-term motions for multi-object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 17939–17948.

- [30] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2018, pp. 839–847.
- [31] L. Jiao, D. Wang, Y. Bai, P. Chen, and F. Liu, "Deep learning in visual tracking: A review," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 54, no. 12, pp. 7780–7792, Dec. 2024.
- [32] Z. Sun, J. Chen, L. Chao, W. Ruan, and M. Mukherjee, "A survey of multiple pedestrian tracking based on tracking-by-detection framework," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 5, pp. 1819–1833, May 2021.
- [33] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 1580–1589.
- [34] Z. Zhang, X. Lu, G. Cao, Y. Yang, L. Jiao, and F. Liu, "VIT-YOLO: Transformer-based YOLO for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 2799–2808.
- [35] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, May 2022.
- [36] Y. Tang, K. Han, J. Guo, C. Xu, C. Xu, and Y. Wang, "GhostNetV2: Enhance cheap operation with long-range attention," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 9969–9982.
- [37] S. Yuan, Y. Wang, T. Liang, W. Jiang, S. Lin, and Z. Zhao, "Real-time recognition and warning of mask wearing based on improved YOLOV5 R6. 1," *Int. J. Intell. Syst.*, vol. 37, no. 11, pp. 9309–9338, 2022.
- [38] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOV7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 7464–7475.
- [39] Y. Zhang et al., "ByteTrack: Multi-object tracking by associating every detection box," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 1–21.
- [40] J. Villacrés, M. Viscaino, J. Delpiano, S. Vougioukas, and F. A. Cheein, "Apple orchard production estimation using deep learning strategies: A comparison of tracking-by-detection algorithms," *Comput. Electron. Agricult.*, vol. 204, Jan. 2023, Art. no. 107513.
- [41] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 658–666.
- [42] F. Yang, S. Odashima, S. Masui, and S. Jiang, "Hard to track objects with irregular motions and similar appearances? Make it easier by buffering the matching space," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2023, pp. 4799–4808.
- [43] W. Wang, K. Zhang, M. Lv, and J. Wang, "Hierarchical spatiotemporal context-aware correlation filters for visual tracking," *IEEE Trans. Cybern.*, vol. 51, no. 12, pp. 6066–6079, Dec. 2021.
- [44] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, 2020, pp. 12993–13000.
- [45] Y. Lan, S. Lin, H. Du, Y. Guo, and X. Deng, "Real-time UAV patrol technology in orchard based on the swin-tyolox lightweight model," *Remote Sens.*, vol. 14, no. 22, p. 5806, 2022.
- [46] X. Zhou, X. Xu, W. Liang, Z. Zeng, and Z. Yan, "Deep-learning-enhanced multitarget detection for end-edge-cloud surveillance in smart IoT," *IEEE Internet Things J.*, vol. 8, no. 16, pp. 12588–12596, Aug. 2021.
- [47] E. Arani, S. Gowda, R. Mukherjee, O. Magdy, S. Kathiresan, and B. Zonooz, "A comprehensive study of real-time object detection networks across multiple domains: A survey," 2023, *arXiv:2208.10895*.
- [48] C. Li et al., "YOLOV6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.
- [49] L. He, X. Liao, W. Liu, X. Liu, P. Cheng, and T. Mei, "FastREID: A pytorch toolbox for general instance re-identification," 2020, *arXiv:2006.02631*.
- [50] S. Lyu, R. Li, Y. Zhao, Z. Li, R. Fan, and S. Liu, "Green citrus detection and counting in orchards based on YOLOV5-CS and AI edge system," *Sensors*, vol. 22, no. 2, p. 576, 2022.
- [51] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 2017, pp. 3645–3649.
- [52] Y. Li, K. Adamczewski, W. Li, S. Gu, R. Timofte, and L. Van Gool, "Revisiting random channel pruning for neural network compression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 191–201.
- [53] Z. Li et al., "When object detection meets knowledge distillation: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 10555–10579, Aug. 2023.
- [54] S. Niu, Y. Liu, J. Wang, and H. Song, "A decade survey of transfer learning (2010–2020)," *IEEE Trans. Artif. Intell.*, vol. 1, no. 2, pp. 151–166, Oct. 2020.
- [55] Y. Wu, H. Sheng, Y. Zhang, S. Wang, Z. Xiong and W. Ke, "Hybrid motion model for multiple object tracking in mobile devices," *IEEE Internet Things J.*, vol. 10, no. 6, pp. 4735–4748, Mar. 2023.
- [56] Y. Zhao, H. Liu, Q. Wang, and C. Li, "A multi-object tracking method using IoU and color fusion model," in *Proc. 9th Int. Conf. Signal Image Process. (ICSIP)*, Nanjing, China, 2024, pp. 747–751, doi: [10.1109/ICSIP61881.2024.10671489](https://doi.org/10.1109/ICSIP61881.2024.10671489).
- [57] Z. Luo, D. Shi, X. Shen, J. Ji and W.-S. Gan, "GFANC-Kalman: Generative fixed-filter active noise control with CNN-Kalman filtering," *IEEE Signal Process. Lett.*, vol. 31, pp. 276–280, 2024.
- [58] A. Balachandran, A. Acharya, S. Chomal, and R. Tharmarasa, "Multipath-assisted multitarget tracking with better track formation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 3, pp. 3021–3037, Jun. 2024.
- [59] L. D. V. Escamilla et al., "Maturity recognition and fruit counting for sweet peppers in greenhouses using deep learning neural networks," *Agriculture*, vol. 14, no. 3, p. 331, 2024.
- [60] Y. Feng et al., "Approach of dynamic tracking and counting for obscured citrus in smart orchard based on machine vision," *Appl. Sci.*, vol. 14, no. 3, p. 1136, 2023, doi: [10.3390/app14031136](https://doi.org/10.3390/app14031136).



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